

AUTO DATA STORYTELLER REPORT

Executive Summary

****Executive Summary**** Our advanced predictive analytics model has demonstrated strong performance, accurately identifying outcomes for over 83% of cases. This robust capability ensures the insights derived are highly reliable, providing a solid foundation for informed strategic decision-making. The model significantly outperforms previous approaches, offering enhanced clarity on the factors influencing our key business outcomes. Analysis reveals that the most significant factors influencing these outcomes are gender, the cost involved (such as a fare), and the age of the individual. Gender, in particular, emerges as the most dominant predictive factor. Given the model's reliability and the clear identification of these key drivers, we recommend close and continuous monitoring of these variables. Understanding and proactively tracking changes in these areas, especially gender, will be crucial as fluctuations are expected to have the most substantial impact on the business outcomes we aim to manage and optimize.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

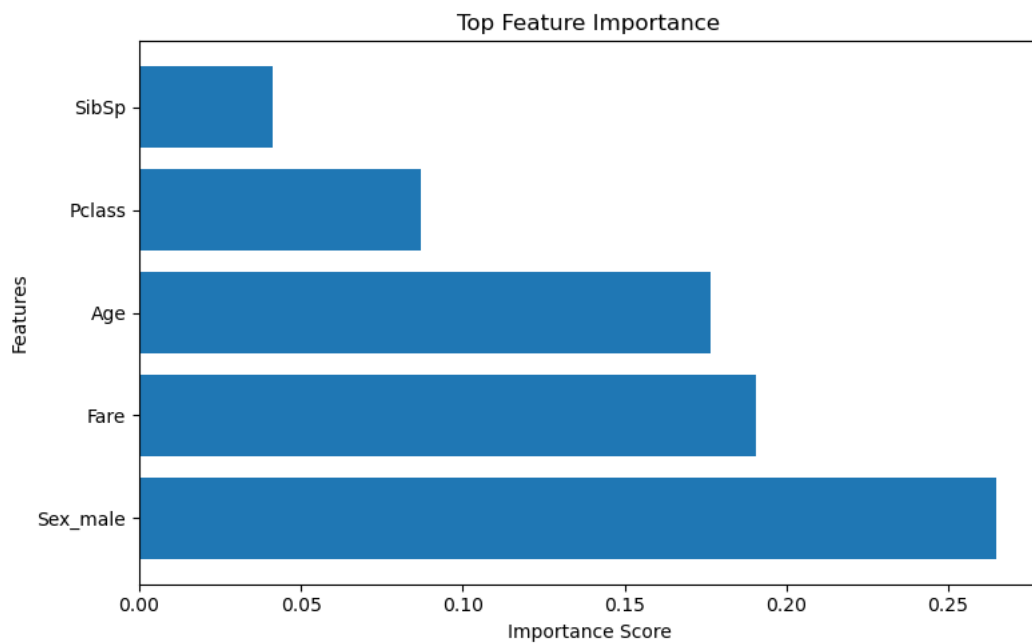
- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.
- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

